Dynamic Modelling & Control of a Multi Environment Quadrotor

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Contents

Introduction

- Aerial-Aquatic Vehicles are designed for diverse environments with potential applications in underwater exploration.
- These vehicles seamlessly transition between aerial and underwater modes, requiring advanced control systems.
- Challenges include:
  - Conflicting design constraints across air and water.
  - Ensuring stability during transitions.
  - Complex dynamics involving morphing capabilities.
- Dynamic model considers:
  - Buoyancy and hydrodynamic characteristics.
  - Propulsion methods and air-water interaction forces.
- The study:
  - Proposes mathematical models and control strategies.
  - Validates stability and robustness via simulations.
  - Demonstrates trajectory tracking in amphibious conditions.

Existing Research

- AUVs and UAVs for amphibious operations are categorized into:
  - Fixed-wing hybrid UAVs (FW-HUAVs).
  - Multi-rotor hybrid UAVs (M-HUAVs).
- Key Contributions:
  - Yang et al. (2013): Fixed-wing amphibious robot with plunge-diving mechanism; prone to impact damage.
  - Drews et al. (2014): First amphibious octa-rotor; limited underwater parameter accuracy.
  - Ranganathan et al. (2015): AQUAD quadrotor; lacked transition strategy.
  - Marcedo et al. (2016): Hybrid vehicle modeled abrupt interface dynamics (e.g., Naviator).

Existing Research

- Common Issues:
  - FW-HUAVs: Stability challenges at the air-water interface.
  - M-HUAVs: Limited endurance, higher weight for dual actuators.
- Control Approaches:
  - Conventional PID: Drews et al. (2014).
  - Advanced Methods: Sliding Mode Control (SMC), Fuzzy PID, Constrained Backstepping.
- Recent Designs:
  - Acutus (2016): Underwater thruster for surge axis control.
  - Looncopter (2016): Ballast tanks for underwater heave actuation.
  - Hydrone: Lateral actuators for efficient underwater motion.
  - Nezha (2019): Combined fixed-wing and underwater glider.

Dynamical Modelling of the Vehicle: Introduction

- Unified dynamic equation for a vehicle in a fluid:

$$\tau_{\text{ext}} = M\dot{v} + C(v)v + D(v)v + g(n)$$

- Where:
  - τ_{ext}: External Forces and Moments
  - $C(v)$: Overall Inertial Matrix (Coriolis and Centripetal term)
  - $D(v)$: Drag term
  - $g(n)$: Restoring force
- Expanded equation considering the environment:

$$\tau_{\text{control}} + \tau_{\text{env}} = (M_{RB} + M_A)\,\dot{v} + (C_{RB}(v) + C_A(v))\,v + D(v)v + g(n)$$

- Assumption: No environmental disturbance during aerial or aquatic phases.
- Key challenge: Differences in medium properties (air vs water) impact terms like Coriolis force, added mass, drag & buoyancy.

Control Forces and Moments (τ_{control})

- Control is done by the propellers and ballast tank (underwater phase).
- Propeller RPM is the control input.
- Propellers are rotated for underwater propulsion, facing forward.
- Control forces can be split into:

$$\tau_{\text{control}} = \tau_{\text{motor}} + \tau_{\text{hull}}$$

- Propulsive thrust force of motors:

$$\tau_{\text{motor}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ K_t\omega_1^2 & K_t\omega_2^2 & K_t\omega_3^2 & K_t\omega_3^2 \\ -K_t\omega_1^2 l & K_t\omega_1^2 l & K_t\omega_1^2 l & -K_t\omega_1^2 l \\ -K_t\omega_1^2 l & -K_t\omega_1^2 l & K_t\omega_1^2 l & K_t\omega_1^2 l \\ K_q\omega_1^2 & K_q\omega_1^2 & K_q\omega_1^2 & K_q\omega_1^2 \end{bmatrix}$$

Overall Mass M

- Overall Mass: Rigidbody Mass and Added Mass.
- Added mass is significant underwater, calculated from the vehicle's volume.
- Pressure hull radius: 0.2 m.
- Added Mass and Inertia:

$$M_a = \frac{4}{6}\pi\rho R^3, \quad I_a = \frac{1}{5}M_a R^2$$

Coriolis Term $C(v)$ - Part 1

- Coriolis term: $C(v) = C_{RB} + C_A$.
- Only significant underwater.
- Coriolis due to rigid body:

$$C_{RB} = \begin{bmatrix} 0 & 0 & 0 & 0 & mw & -mv \\ 0 & 0 & 0 & -mw & 0 & mu \\ 0 & 0 & 0 & mv & -mu & 0 \\ 0 & -mw & mv & 0 & I_{zz}r & -I_{yy}q \\ mw & 0 & -mu & -I_{zz}r & 0 & I_{xx}p \\ -mv & mu & 0 & I_{yy}q & -I_{xx}p & 0 \end{bmatrix}$$

Coriolis Term $C(v)$ - Part 2

- Coriolis due to added mass:

$$C_A = \begin{bmatrix} 0 & 0 & 0 & 0 & -a_3 & a_2 \\ 0 & 0 & 0 & a_3 & 0 & -a_1 \\ 0 & 0 & 0 & -a_2 & a_1 & 0 \\ 0 & -a_3 & a_2 & 0 & -b_3 & b_2 \\ a_3 & 0 & -a_1 & b_3 & 0 & -b_1 \\ -a_2 & a_1 & 0 & -b_2 & b_1 & 0 \end{bmatrix}$$

- Expressions for a_1, a_2, a_3, b_1, b_2, and b_3:

$$a_1 = X_{\dot{u}}u + X_{\dot{v}}v + X_{\dot{w}}w + X_{\dot{p}}p + X_{\dot{q}}q + X_{\dot{r}}r$$

$$a_2 = Y_{\dot{u}}u + Y_{\dot{v}}v + Y_{\dot{w}}w + Y_{\dot{p}}p + Y_{\dot{q}}q + Y_{\dot{r}}r$$

$$a_3 = Z_{\dot{u}}u + Z_{\dot{v}}v + Z_{\dot{w}}w + Z_{\dot{p}}p + Z_{\dot{q}}q + Z_{\dot{r}}r$$

$$b_1 = K_{\dot{u}}u + K_{\dot{v}}v + K_{\dot{w}}w + K_{\dot{p}}p + K_{\dot{q}}q + K_{\dot{r}}r$$

$$b_2 = M_{\dot{u}}u + M_{\dot{v}}v + M_{\dot{w}}w + N_{\dot{p}}p + M_{\dot{q}}q + M_{\dot{r}}r$$

$$b_3 = N_{\dot{u}}u + N_{\dot{v}}v + N_{\dot{w}}w + N_{\dot{p}}p + N_{\dot{q}}q + N_{\dot{r}}r$$

Drag Term $D(v)$ and Restoring Term $g(n)$

- **Drag Term:** Friction due to fluid interaction.
- Significant underwater for control.
- Drag expression:

$$\tau_{\text{drag}} = -\frac{1}{2} C_d A_r v|v|$$

- **Restoring Term:** Restoring force due to buoyancy.
- Significant underwater (air's low density).
- Restoring force expression:

$$g(n) = \begin{bmatrix} F_g - F_b \\ (r_g \times F_g) - (r_b \times F_b) \end{bmatrix}$$

Transition Strategy

- Medium-dependent parameters experience abrupt changes during the transition.
- These parameters can be approximated using a linear polynomial, depending on the transition height h.
- Critical layer thickness h defines the air-water interface:
  - Center of gravity (C.G) is at $(0, 0, -h)$ when the robot is in air.
  - C.G moves to $(0, 0, +h)$ when fully submerged in water.
- Medium-dependent parameters (drag, thrust, torque coefficients) vary linearly within the transition layer ($Z = -h$ to $Z = h$).

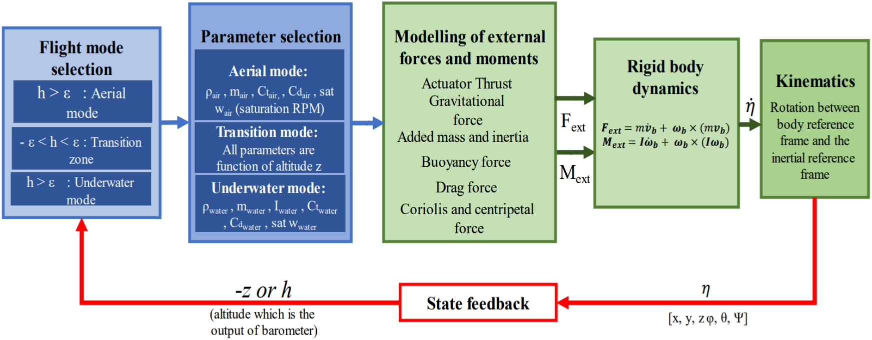
Transition Strategy

- Fluid density is modeled as a piecewise function:

$$\rho = \begin{cases} \rho_1, & z \leq -h \\ 0.5 \times \left(\rho_1 \cdot (1 - \frac{z}{h}) + \rho_2 \cdot (1 + \frac{z}{h})\right), & -h \leq z \leq h \\ \rho_2, & z \geq h \end{cases}$$

- Fluid density varies linearly within the transition layer based on the inertial position of the robot.
- This strategy is applied to other parameters like drag and thrust coefficients.
- These parameters are fed into the dynamic model to simulate robot behavior.

Schematic for Dynamic Modelling



Aerial Controller

- Controls (x, y, z, ψ).
- Issues with large errors in x, y:
  - Risk of tipping over due to high acceleration.
- Solution:
  - Stacked PID controllers.
  - Attitude PID for controlling tilt angles.
  - Altitude PID for maintaining height.

Aerial Motor Mixing Algorithm

- Takes propeller positions and frame of reference into account.
- Frame-dependent algorithm for calculating motor speeds.

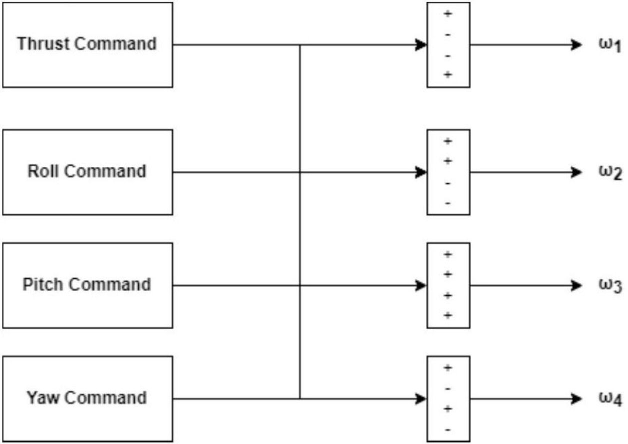


Figure: Aerial Motor Mixing Algorithm

x Controller Strategy

- Calculates pitch angle for x-axis control:
  - Errors: x_{ref} (desired) vs. x_e (current).
  - PID controller for pitch angle.
  - Saturation to avoid over-tilting.
- Outputs θ_{ref} to θ Controller Strategy.

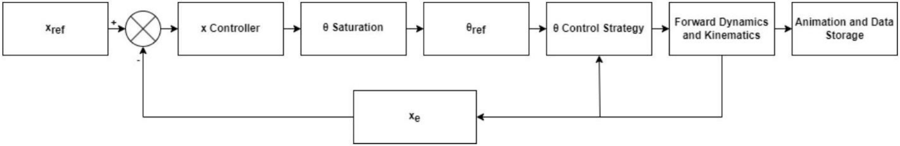


Figure: x Controller Strategy

θ Controller Strategy

- Input:
  - Pitch error from dynamics block.
  - Reference pitch θ_{ref}.
- Calculates pitch command for motor speeds.

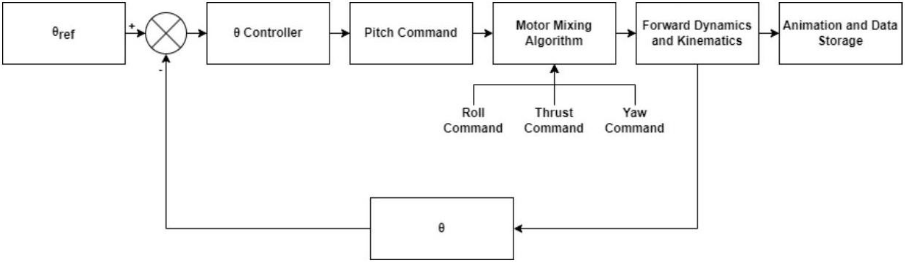


Figure: θ Controller Strategy

y Controller Strategy

- Calculates roll angle for y-axis control:
  - Errors: y_{ref} (desired) vs. y_e (current).
  - PID controller for roll angle.
  - Saturation to avoid over-tilting.
- Outputs ϕ_{ref} to ϕ Controller Strategy.

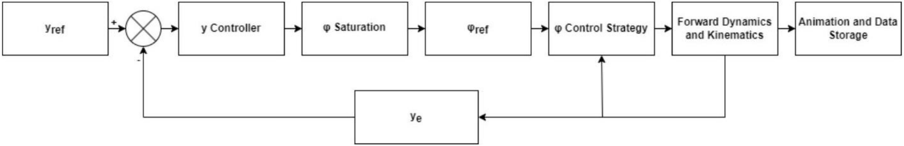


Figure: y Controller Strategy

ϕ Controller Strategy

- Input:
  - Roll error from dynamics block.
  - Reference roll ϕ_{ref}.
- Calculates roll command for motor speeds.

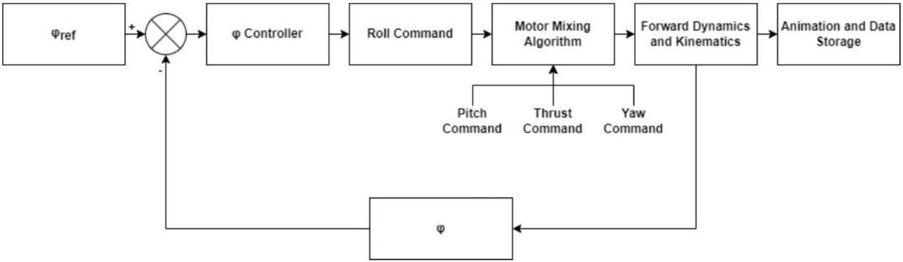


Figure: ϕ Controller Strategy

z Controller Strategy

- Modified PID controller for altitude control:
  - Maintains stable hover RPM.
  - Errors: z_{ref} (desired) vs. z_e (current).
- Calculates thrust command for motor speeds.

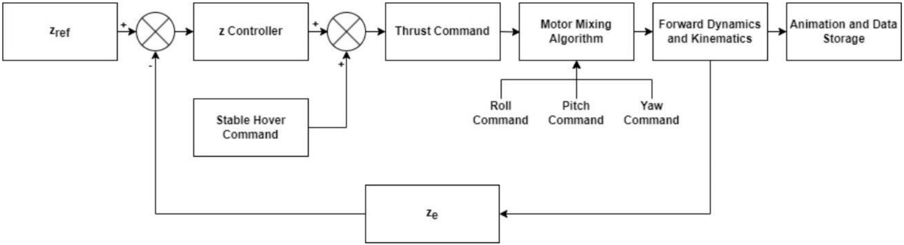


Figure: z Controller Strategy

ψ Controller Strategy

- Controls yaw:
  - Yaw error from dynamics block.
  - PID controller for yaw angle.
  - Reference yaw ψ_{ref}.
- Calculates yaw command for motor speeds.

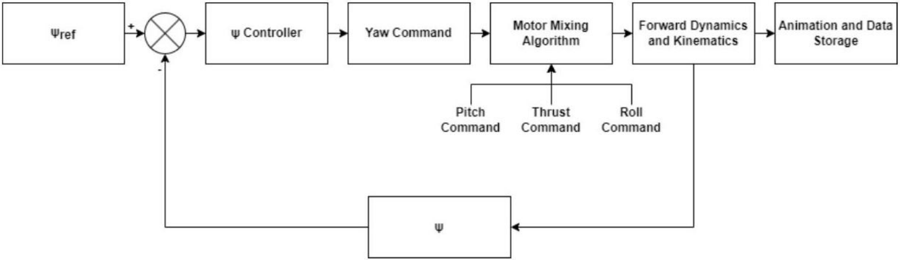


Figure: ψ Controller Strategy

Aquatic Controller

- Propellers turn -90° in the y-axis during the transition phase.
- Changes in Forces and Moments due to reoriented propellers.
- Medium properties:
  - Increased density.
  - Significant Coriolis, Drag, and Added Mass terms.
- RPMs limited to 250 RPM to avoid mechanical stress.
- Aquatic control strategy transforms control from (x, y, z, ψ) to (r, z, ψ) in cylindrical coordinates.

Coordinate Transformation Formula:

$$r = \sqrt{(x_{\text{ref}} - x_e)^2 + (y_{\text{ref}} - y_e)^2}$$

$$\psi = \tan^{-1}\left(\frac{y_{\text{ref}} - y_e}{x_{\text{ref}} - x_e}\right)$$

Aquatic Motor Mixing Algorithm

- Algorithm accounts for propeller positions and frame of reference.
- Key feature: Thrust Command moves vehicle forward instead of vertically.

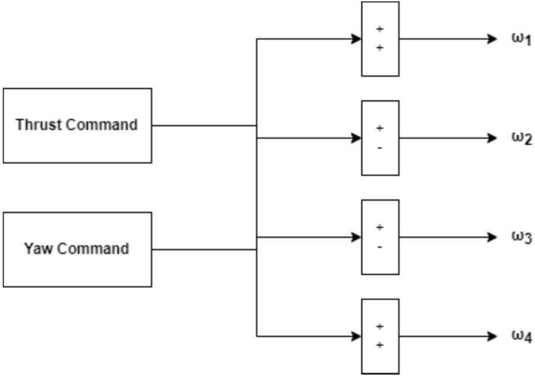


Figure: Aquatic Motor Mixing Algorithm

r Controller Strategy

- Input: Magnitude of horizontal distance between reference and current positions.
- Output: Thrust Command to Motor Mixing Algorithm Block.
- Motor RPMs set to 0 within a threshold radius of the goal position.

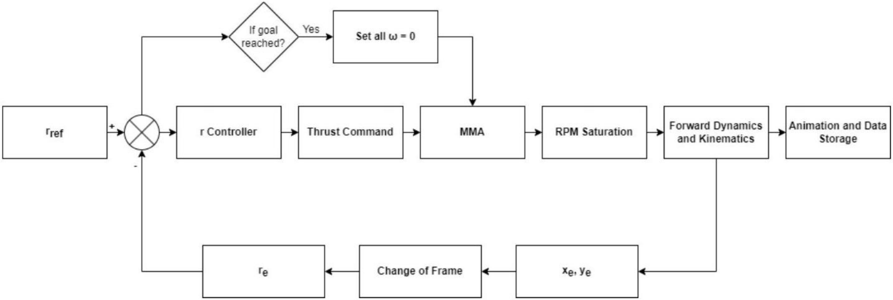


Figure: r Controller Strategy

ψ Controller Strategy

- Input: Minimum absolute difference between reference yaw and current yaw.
- Output: Yaw Command to Motor Mixing Algorithm Block.

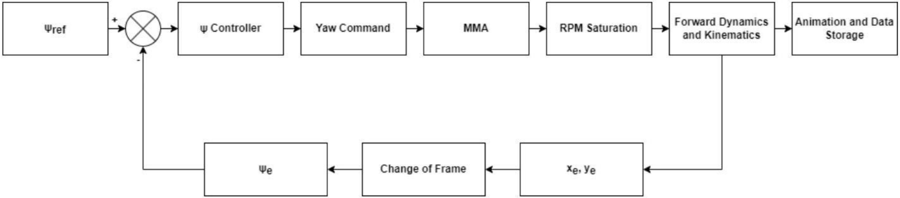


Figure: ψ Controller Strategy

z Controller Strategy

- Input: Difference between reference depth and current depth.
- Output: Modified thrust accounting for added mass, water weight, and vehicle mass.
- Two ballast tank intakes water based on the following condition:

$$Mass_{water} = \begin{cases} \frac{2\rho l \pi d^2}{4} & \text{if } Z_e > 0 \\ 0 & \text{otherwise} \end{cases}$$

where ρ is the water density, l is the length, and d is the diameter of the ballast tank.

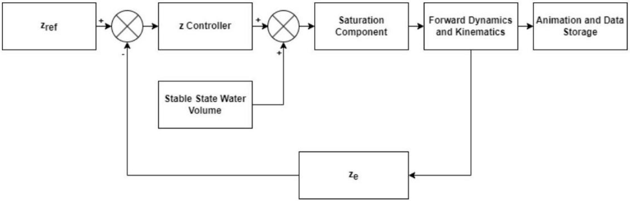


Figure: z Controller Strategy

Transition Control Strategy

- During the transition:
  - All rotors stop rotating when transitioning from air to water.
  - A thrust of $\frac{w}{10}$ is applied to all rotors when transitioning from water to air upon crossing the $z = 0$ line.
- Rotor rotation happens fully in the aquatic environment between $z = \frac{h}{2}$ and $z = 0$ where θ varies linearly, the rotation matrix:

$$R_\theta = \begin{bmatrix} \cos(\theta) & 0 & -\sin(\theta) & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ \sin(\theta) & 0 & \cos(\theta) & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

Simulation: Introduction

- To verify the working of the proposed controller, we will perform following 2 tests for each separate environment:
  - Position control.
  - Trajectory tracking.
- Simulations performed for the Quadrotor is in the MATLAB Simulink environment.
- Due to Computational constraints, the simulation is ran in the respective environments, but cannot visualized as then each of simulation takes more than an hour to run.
- Detailed simulation parameters are tabulated, as shown in the following table.

Physical Parameters used in the Simulation

Parameter	Value used in Simulation
Configuration	Quad-copter ‘X’ configuration
Overall weight / length / breadth	2.0 kg / 1.0 m / 1.0 m
Overall volume	0.00419 m^3
Pressure hull dimensions	Sphere (Diameter: 0.20 m)
Centre of gravity	[0, 0, 0], Origin of body
Centre of Buoyancy	Depends on the height of water
Density of Air/ Water	1.225 kg/m^3 / 1000 kg/m^3
Thrust coefficient Air / Water	1.5e-5 / 0.0392
Torque coefficient Air / Water	4.2e-6 / 0.0470
Drag coefficient	[0.82, 0.82 0.47]
Drag Platform Area	[0.2, 0.2, 0.069] m^2
Inertial terms	[0.00317 kgm^2, 0.00317 kgm^2, 0.0704 kgm^2]

Table 4.1: Physical Parameters used in the Simulation

Environment Visualization

- As mentioned above, since we cannot visualize the environment while running the simulation, here is the visualization of the air, water & land environments.
- So, all the simulations ran in the results have been visualized by graphs only, where $z > 0$ is aquatic & $z < 0$ is aerial environment.

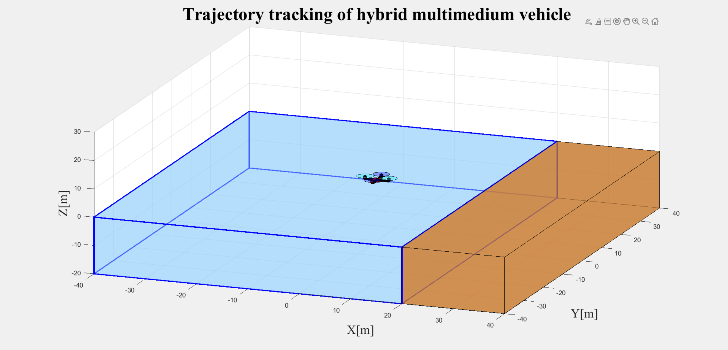


Figure: Visualization of the Environment Coded in Simulink

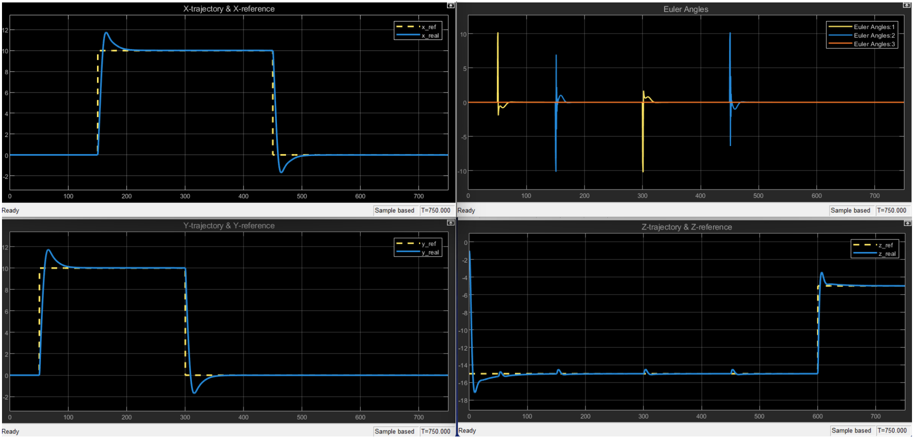


Figure: Position Control in Aerial Environment

Trajectory Control in Aerial Environment

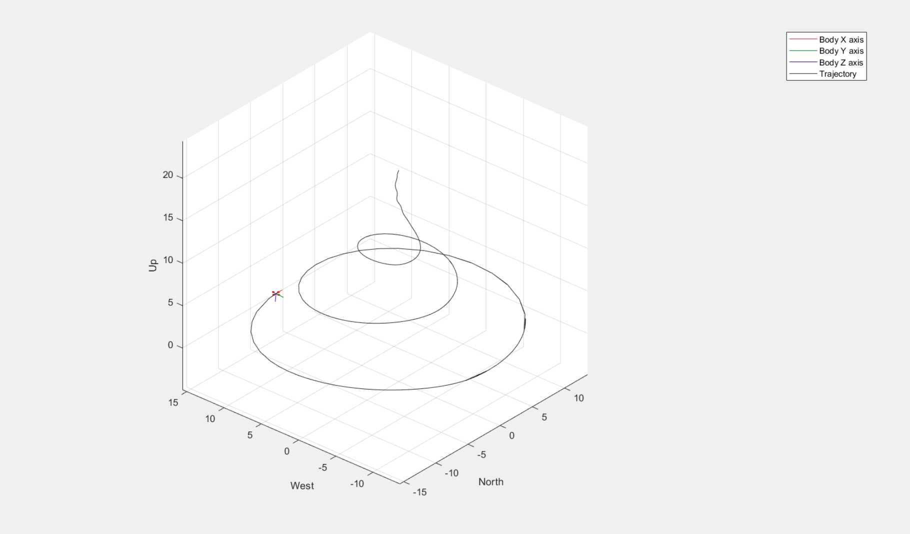


Figure: Trajectory Control in Aerial Environment

Trajectory Control in Aerial Environment

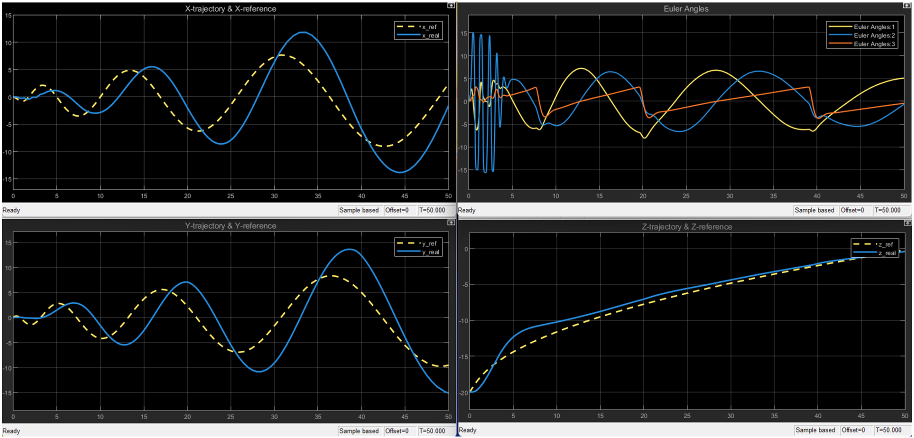


Figure: Trajectory Control in Aerial Environment

A few Other Trajectories Generated in Aerial Environment

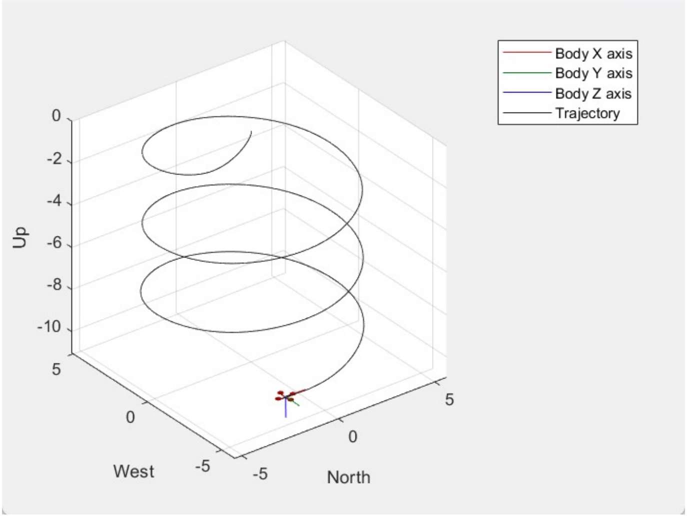


Figure: Trajectory Control in Aerial Environment

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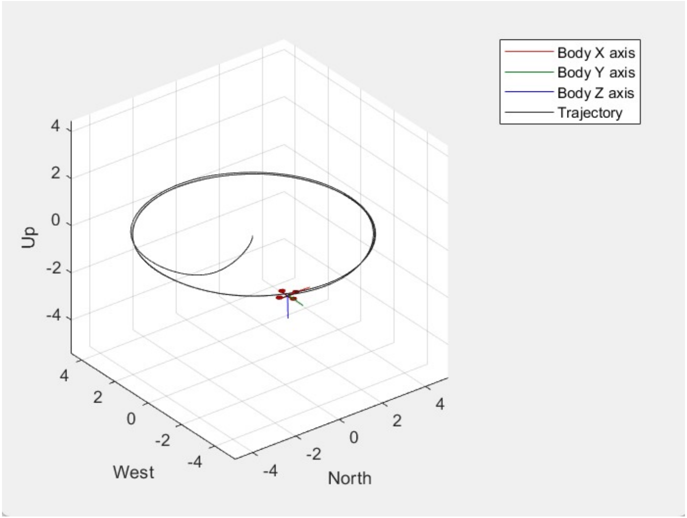

Figure: Trajectory Control in Aerial Environment

A few Other Trajectories Generated in Aerial Environment

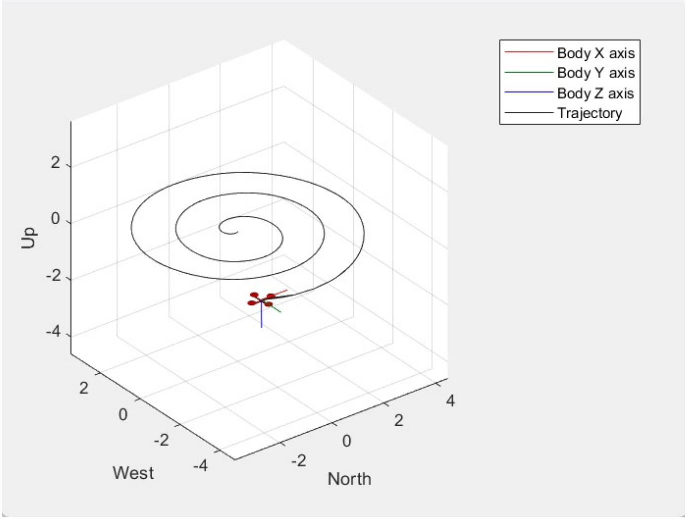

Figure: Trajectory Control in Aerial Environment

Position Control in Aquatic Environment

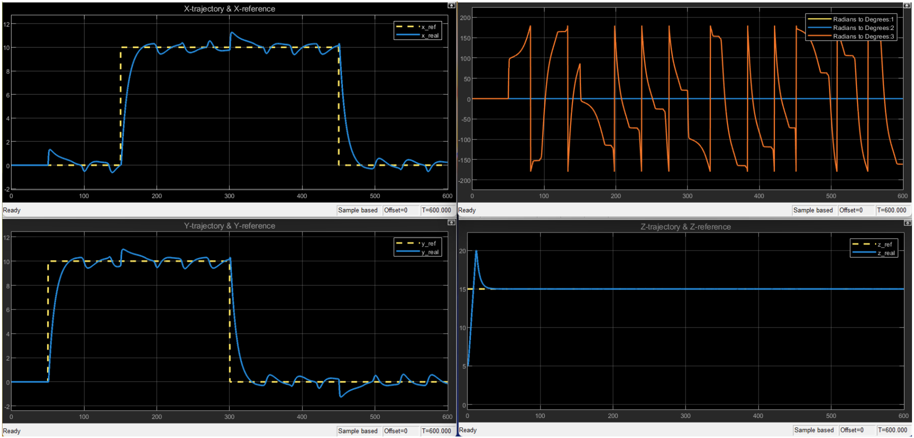

Figure: Position Control in Aquatic Environment

Trajectory Control in Aquatic Environment

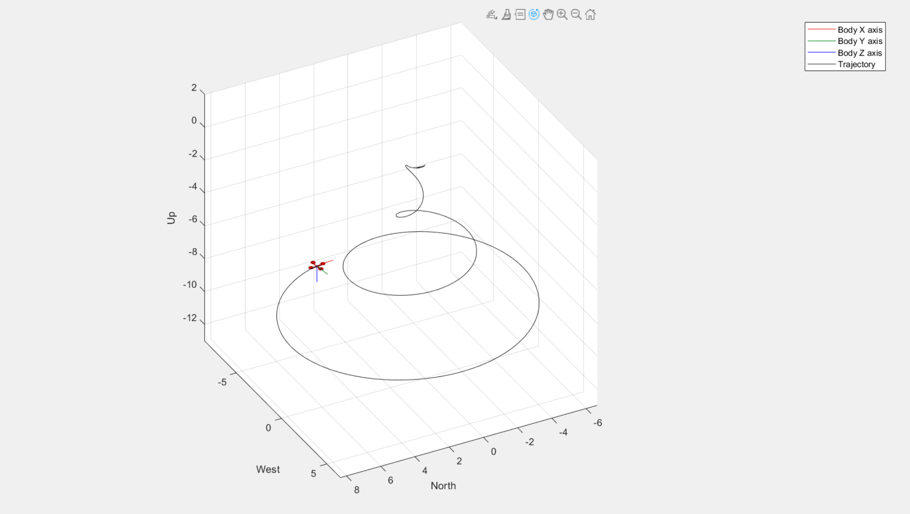

Figure: Trajectory Control in Aquatic Environment

Trajectory Control in Aquatic Environment

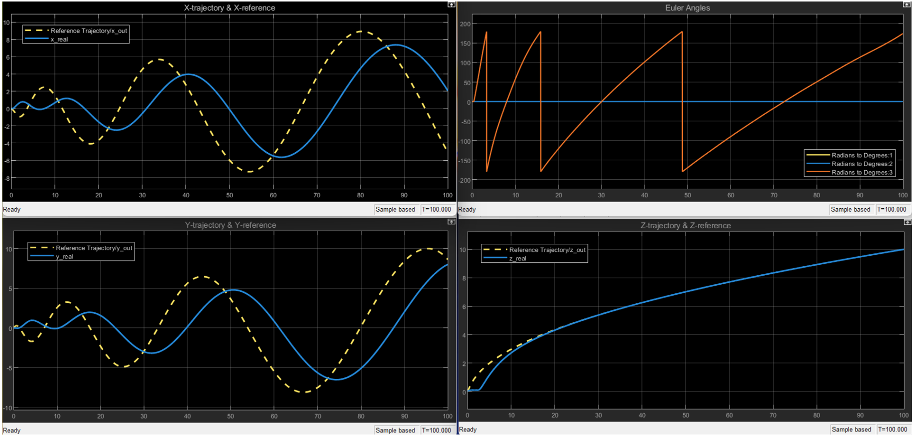

Figure: Trajectory Control in Aquatic Environment

Position Control in Aerial-Aquatic Environment

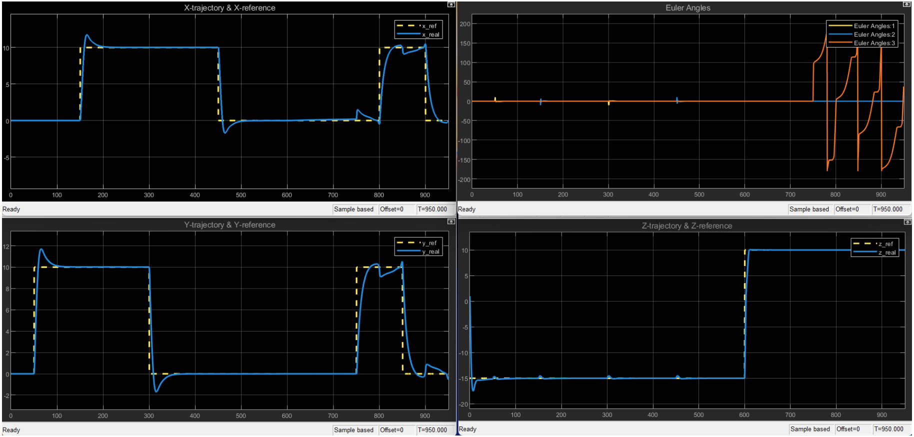

Figure: Position Control in Aerial-Aquatic Environment

Trajectory Control in Aerial-Aquatic Environment

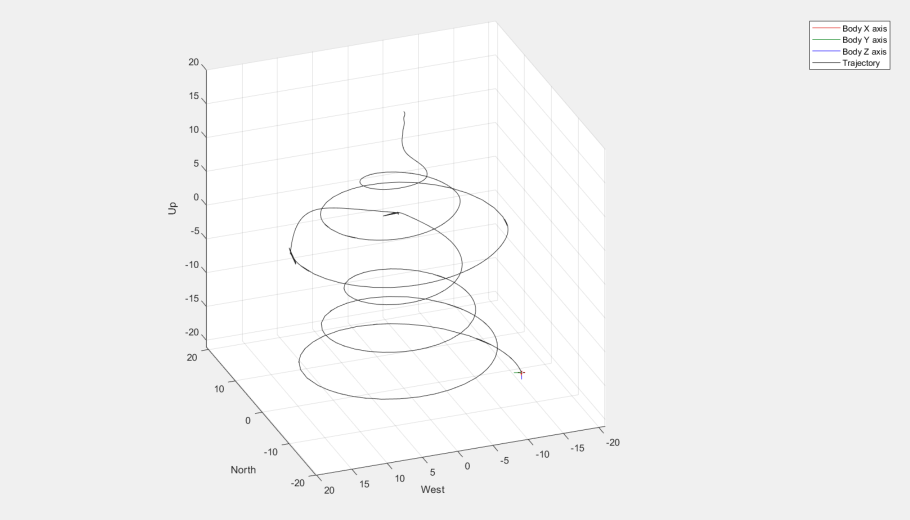

Figure: Trajectory Control in Aerial-Aquatic Environment

Trajectory Control in Aerial-Aquatic Environment

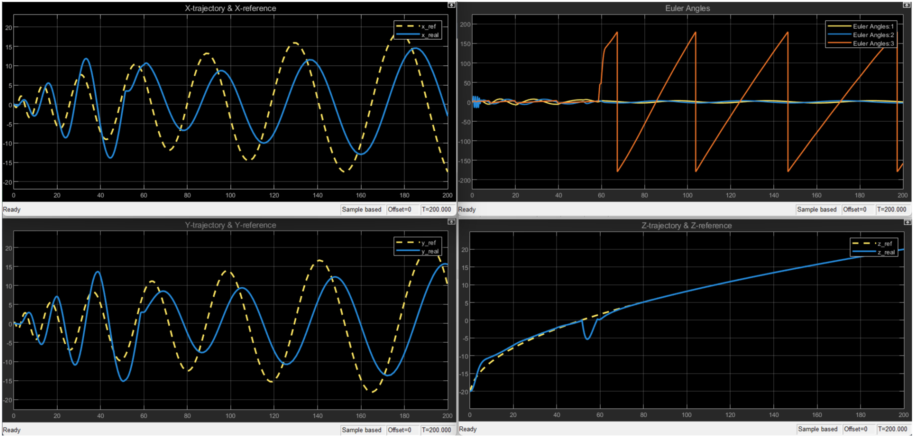

Figure: Trajectory Control in Aerial-Aquatic Environment

Ongoing Work

- Fixing the aquatic x-y controller:
  - Addressing steady-state fluctuations.
  - Improving controller performance to achieve stability.
- Implementing developed control algorithms into a real drone for real-world testing.
- Modeling dynamics and control for terrestrial locomotion.

Ongoing Work: Terrestrial Environment Dynamics

- Modeling dynamics and control for terrestrial locomotion.
- Reference frames used:
  - Inertial frame I with $\hat{i}_3$ opposite to the gravity vector.
  - Frame E fixed at the cage's center with $\hat{e}_3$ aligned with $\hat{i}_3$.
  - Frame C fixed to the cage with $\hat{c}_2$ pointing along $\hat{e}_2$.
  - Quadrotor-fixed frame B with $\hat{b}_3$ normal to the quadrotor, pointing upward.
- During terrestrial locomotion:
  - Propeller thrust rolls the cage on the ground.
  - Overcoming rolling resistance, turning resistance, and air drag.
- Rolling resistance torque:

$$\tau_r = C_{rr} R ||N|| \hat{b}_2$$

- Normal force:

$$N = Mg - u_f \cos(\theta) \hat{i}_3, \quad M = m_B + m_C$$

Drone in Terrestrial Environment

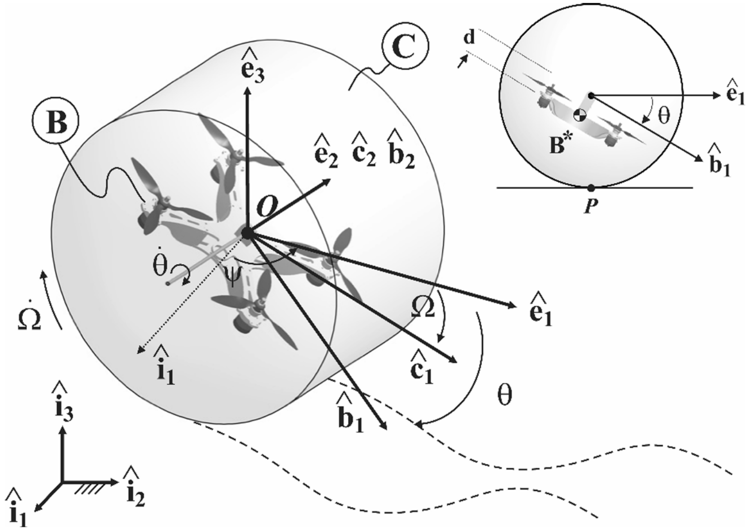

Figure: Drone in Terrestrial Environment

References

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